

3/4 UNIT MATHEMATICS FORM VI

Time allowed: 2 hours (plus 5 minutes reading)

Exam date: 8th August, 2000

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the left margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- Each question will be collected separately.
- Start each question in a new 8-leaf answer booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each answer booklet.

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QUESTION ONE (Start a new answer booklet)

Marks

1 (a) Differentiate $\tan^{-1}(\pi x)$ with respect to x .

2 (b) Find:

(i) $\int \frac{dx}{4+x^2},$

(ii) $\int \frac{dx}{\sqrt{4-x^2}}.$

3 (c) Evaluate $\int_0^{\frac{\pi}{4}} \tan^2 x \, dx.$

2 (d) Find the value of $\tan \alpha$ if α is the acute angle between the lines $y = \frac{1}{2}x$ and $y = -\frac{1}{\sqrt{3}}x + 1.$

2 (e) Prove that $\frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \sec \theta.$

2 (f) Solve the inequation $\frac{x-2}{x} \geq 1.$

QUESTION TWO (Start a new answer booklet)

Marks

3 (a) Use the substitution $u = e^x$ to evaluate $\int_0^{\ln \sqrt{3}} \frac{e^x}{1+e^{2x}} \, dx.$

3 (b) Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6.$

3 (c) The volume V of a spherical balloon is expanding at the rate of $10 \text{ mm}^3/\text{s}$. Find the rate of increase of its radius r when the surface area S is 1000 mm^2 . (Note: You may use the formulae $V = \frac{4}{3}\pi r^3$ and $S = 4\pi r^2$).

3 (d) (i) Sketch, without the use of calculus, the polynomial $P(x) = (2x-1)^2(x+1)^3$, showing the x - and y -intercepts.
(ii) Hence solve the inequation $P(x) \geq 0.$

QUESTION THREE (Start a new answer booklet)

Marks

- 4 (a) The velocity v of a particle moving along the x -axis satisfies the equation

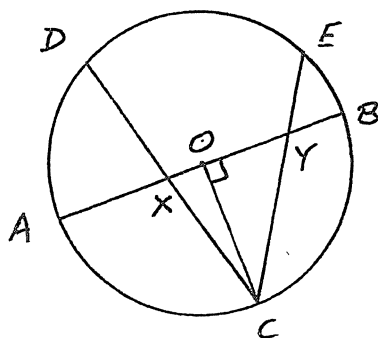
$$v^2 = 5 + 14x - 3x^2.$$

Show that the particle is moving in simple harmonic motion, and find the centre, amplitude and period of the motion.

- 4 (b) (i) Express $\cos 2x - \sin 2x$ in the form $R \cos(2x + \alpha)$, where α is acute and $R > 0$.

(ii) Hence solve the equation $\cos 2x - \sin 2x = 1$, for $0 \leq x \leq \pi$.

- 4 (c)



In the diagram above, AB is the diameter of a circle with centre O . The radius OC is drawn perpendicular to AB . The chords CD and CE intersect the diameter in the points X and Y respectively.

Copy the diagram into your examination booklet.

- (i) Prove that $\angle CBA = \angle CAB = 45^\circ$.
- (ii) Give a reason why $\angle DCA = \angle DBA$ and $\angle CBD = \angle CED$.
- (iii) Prove that $\angle CBD = \angle CXB$. (Hint: Let $\angle DCA = \alpha$).
- (iv) Prove that $XYED$ is a cyclic quadrilateral.

QUESTION FOUR (Start a new answer booklet)

Marks

- 4 (a) Prove by mathematical induction that for all positive integers n ,

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n! = (n+1)! - 1.$$

- 4 (b) The variable point P has coordinates $P(a \cos 2\theta, a \sin \theta)$.

(i) Show that P lies on the curve $y^2 = -\frac{a}{2}(x - a)$.

(ii) Sketch the locus of P as θ varies, taking account of any restrictions on x and y .

- 4 (c) The point $P(x, y)$ divides the interval joining the points $A(-1, 3)$ and $B(2, 8)$ internally in the ratio $k : 1$.
- (i) Find the coordinates of P in terms of k .
- (ii) Hence find the ratio in which the line $5x + 2y - 10 = 0$ divides the interval AB .

QUESTION FIVE (Start a new answer booklet)

Marks

- 4 (a) When the polynomial $P(x)$ is divided by $(x + 1)(x - 2)$, the result can be written as

$$P(x) = (x + 1)(x - 2)Q(x) + R(x), \text{ where } R(x) = ax + b.$$

- (i) Given that $P(-1) = 3$, find the value of $R(-1)$.
- (ii) Given also that the remainder is -2 when $P(x)$ is divided by $x - 2$, find the values of a and b .

- 4 (b) Let the expansion of $(2 + 3x)^{12}$ be written in the form $\sum_{r=0}^{12} t_r x^r$.

- (i) Write down expressions for t_r and t_{r+1} , and show that $\frac{t_{r+1}}{t_r} = \frac{36 - 3r}{2r + 2}$.
- (ii) Hence find the greatest coefficient in the expansion of $(2 + 3x)^{12}$. You need not simplify your answer.

- 4 (c) (i) Show that the coefficient of x^n in the expansion of $(1 + x)^n(1 + x)^n$ is given by

$$\sum_{r=0}^n ({}^nC_r)^2.$$

- (ii) Hence, by equating the coefficients of x^n on both sides of the identity

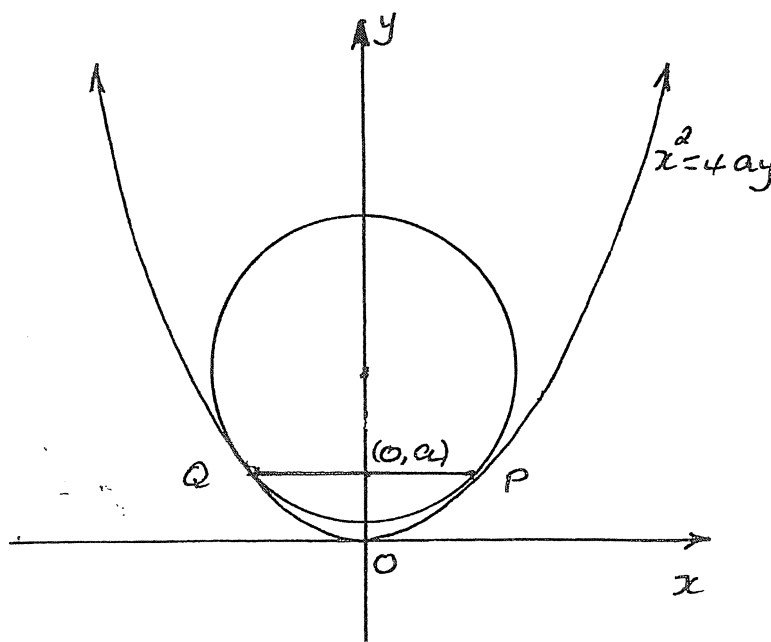
$$(1 + x)^n(1 + x)^n = (1 + x)^{2n},$$

prove that $\sum_{r=0}^n ({}^nC_r)^2 = \frac{(2n)!}{(n!)^2}.$

QUESTION SIX (Start a new answer booklet)

Marks

- 5** (a) (i) Show that the function $f(x) = x^3 - 3x + 1$ has stationary points at $x = 1$ and at $x = -1$.
- (ii) Show that $x^3 - 3x + 1 = 0$ has a root α between $x = 0$ and $x = 0.5$.
- (iii) Taking $x = 0.1$ as a first approximation, use one application of Newton's method to find a closer approximation to α . Give your answer correct to three decimal places.
- (iv) Explain, with the aid of a neat sketch of the curve, why $x = 1.1$ would not be a suitable first approximation to α .
- 7** (b) (i) Show that the equation of the chord PQ joining the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ on the parabola $x^2 = 4ay$ is $y = \frac{1}{2}(p+q)x - apq$.
- (ii) Suppose now that PQ is the *latus rectum* of the parabola, that is, the chord parallel to the directrix passing through the focus.
- (α) Show that $p + q = 0$ and $pq = -1$, and find the coordinates of P and Q .
- (β) Use calculus to find the equations of the normals at P and Q , and show that they intersect at $N(0, 3a)$.
- (γ)



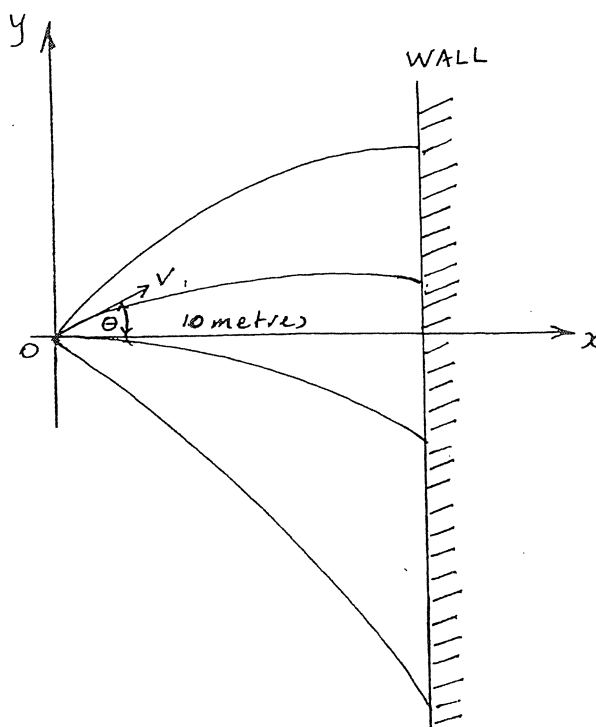
The diagram above shows the circle that touches the parabola $x^2 = 4ay$ at the endpoints of the latus rectum. Use (β) above to prove that the equation of the circle is $x^2 + y^2 - 6ay + a^2 = 0$.

QUESTION SEVEN (Start a new answer booklet)

Marks

3 (a) Find the general solution of the equation $\cos\left(2\pi\left(1 - \frac{1}{3}x\right)\right) = -\frac{1}{2}$.

9 (b)



In the diagram above, a large number of projectiles are fired simultaneously from O , each with the same velocity V but various angles of elevation θ , at a wall distant 10 metres from O . The projectiles are fired so that their trajectories all lie in the same vertical plane perpendicular to the wall.

You may assume that the equations for the coordinates of a projectile at time t are

$$x = Vt \cos \theta \text{ and } y = -\frac{1}{2}gt^2 + Vt \sin \theta.$$

- (i) Use the identity $\cos^2 \theta + \sin^2 \theta = 1$ to eliminate θ from these two equations, and hence prove that the relationship between height y and time t is

$$4y^2 + 4gt^2y + k = 0, \text{ where } k = g^2t^4 + 4x^2 - 4V^2t^2.$$

- (ii) Show that the first impact on the wall occurs at time $t = \frac{10}{V}$, and that this projectile was fired horizontally. Also find where this projectile hits the wall.

- (iii) Show that for $t > \frac{10}{V}$, there are two impacts at time t , and that the distance between these impacts is

$$2\sqrt{V^2t^2 - 100}.$$

- (iv) Given that $V = 10$ m/s, what are the initial angles of elevation of the two projectiles that strike the wall simultaneously $20\sqrt{3}$ metres apart.

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QUESTION ONE

$$(a) \frac{d}{dx}(\tan^{-1} \pi x) = \frac{\pi}{1 + \pi^2 x^2}. \quad \boxed{\checkmark}$$

$$(b) \quad (i) \int \frac{dx}{4 + x^2} = \frac{1}{2} \tan^{-1} \frac{x}{2} + C. \quad \boxed{\checkmark}$$

$$(ii) \int \frac{dx}{\sqrt{4 - x^2}} = \sin^{-1} \frac{x}{2} + C. \quad \boxed{\checkmark}$$

$$\begin{aligned} (c) \int_0^{\frac{\pi}{4}} \tan^2 x \, dx &= \int_0^{\frac{\pi}{4}} (\sec^2 x - 1) \, dx, \quad \boxed{\checkmark} \\ &= \left[\tan x - x \right]_0^{\frac{\pi}{4}} \quad \boxed{\checkmark} \\ &= 1 - \frac{\pi}{4} \\ &= \frac{4 - \pi}{4}. \quad \boxed{\checkmark} \end{aligned}$$

$$(d) \text{ Gradients are } \frac{1}{2} \text{ and } -\frac{1}{\sqrt{3}}.$$

$$\begin{aligned} \tan \alpha &= \left| \frac{\frac{1}{2} + \frac{1}{\sqrt{3}}}{1 - \frac{1}{2\sqrt{3}}} \right| \\ &= \frac{\sqrt{3} + 2}{2\sqrt{3} - 1}. \quad \boxed{\checkmark\checkmark} \end{aligned}$$

$$\begin{aligned} (e) \, LHS &= \frac{\sin 2\theta \cos \theta - \cos 2\theta \sin \theta}{\sin \theta \cos \theta} \quad \boxed{\checkmark} \\ &= \frac{\sin \theta}{\sin \theta \cos \theta} \\ &= \sec \theta \quad \boxed{\checkmark} \\ &= RHS. \end{aligned}$$

$$(f) \quad \frac{x-2}{x} \geq 1$$

$$(x-2)x \geq x^2, \quad x \neq 0 \quad \boxed{\checkmark}$$

$$x^2 - 2x \geq x^2$$

$$-2x \geq 0$$

$$\text{so} \quad x \leq 0 \quad \boxed{\checkmark}$$

QUESTION TWO

(a) Let $u = e^x$

$$du = e^x dx.$$

When $x = 0$

$$u = 1.$$

When $x = \ln\sqrt{3}$

$$u = \sqrt{3}.$$

$$\begin{aligned} \int_0^{\ln\sqrt{3}} \frac{e^x}{1+e^{2x}} dx &= \int_1^{\sqrt{3}} \frac{du}{1+u^2} \quad \boxed{\checkmark} \\ &= \left[\tan^{-1} u \right]_1^{\sqrt{3}} \quad \boxed{\checkmark} \\ &= \frac{\pi}{3} - \frac{\pi}{4} \\ &= \frac{\pi}{12}. \quad \boxed{\checkmark} \end{aligned}$$

(b) General term $= {}^6C_r (x^2)^{6-r} (2x^{-1})^r$
 $= {}^6C_r \times 2^r \times x^{12-3r} \quad \boxed{\checkmark}$

$$12 - 3r = 0$$

$$r = 4. \quad \boxed{\checkmark}$$

Required term $= {}^6C_4 \times 2^4$
 $= 240. \quad \boxed{\checkmark}$

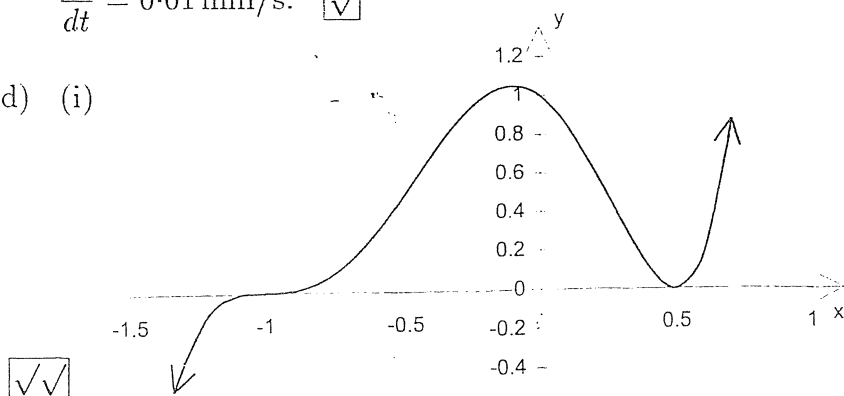
(c) $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$

$$10 = 4\pi r^2 \times \frac{dr}{dt} \quad \boxed{\checkmark}$$

$$10 = 1000 \times \frac{dr}{dt} \quad \boxed{\checkmark}$$

$$\frac{dr}{dt} = 0.01 \text{ mm/s.} \quad \boxed{\checkmark}$$

(d) (i)



(ii) $x \geq -1. \quad \boxed{\checkmark}$

QUESTION THREE

$$(a) \quad v^2 = 5 + 14x - 3x^2$$

$$\frac{d}{dx}(\frac{1}{2}v^2) = \frac{1}{2}(14 - 6x)$$

$$= 7 - 3x$$

$$= -3(x - \frac{7}{3}), \quad \boxed{\checkmark}$$

which is of the required form.

$$\text{The centre of motion is } x = \frac{7}{3}. \quad \boxed{\checkmark}$$

$$\text{Let } v^2 = 0$$

$$3x^2 - 14x - 5 = 0$$

$$(3x + 1)(x - 5) = 0$$

$$x = -\frac{1}{3} \text{ or } 5.$$

$$\text{Amplitude} = 5 - \frac{7}{3}$$

$$= \frac{8}{3}. \quad \boxed{\checkmark}$$

$$n = \sqrt{3}$$

$$\text{Period} = \frac{2\pi}{\sqrt{3}}. \quad \boxed{\checkmark}$$

$$(b) \quad (i) \quad \text{Let } \cos 2x - \sin 2x = R \cos(2x + \alpha)$$

$$= R \cos 2x \cos \alpha - R \sin 2x \sin \alpha$$

$$\text{So } R \cos \alpha = 1$$

$$\text{and } R \sin \alpha = 1.$$

$$\text{So } R = \sqrt{2}$$

$$\text{and } \alpha = \frac{\pi}{4}.$$

$$\text{So } \cos 2x - \sin 2x = \sqrt{2} \cos(2x + \frac{\pi}{4}). \quad \boxed{\checkmark}$$

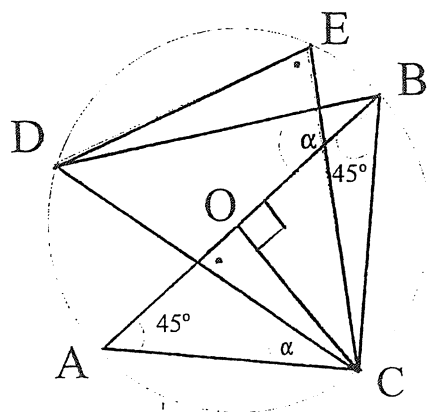
$$(ii) \quad \text{Now } \sqrt{2} \cos(2x + \frac{\pi}{4}) = 1$$

$$\cos(2x + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}, \text{ for } \frac{\pi}{4} \leq 2x + \frac{\pi}{4} \leq \frac{9\pi}{4} \quad \boxed{\checkmark}$$

$$2x + \frac{\pi}{4} = \frac{\pi}{4}, \frac{7\pi}{4}, \text{ or } \frac{9\pi}{4}$$

$$x = 0, \frac{3\pi}{4} \text{ or } \pi. \quad \boxed{\checkmark\checkmark}$$

(c)



(i) $\angle CBA = \angle CAB = 45^\circ$ (angle at centre is twice angle at circumference). $\square$

(ii) $\angle DCA = \angle DBA$ and $\angle CBD = \angle CED$,
(angles at the circumference standing on the same arc). $\square$

(iii) Let $\angle DCA = \angle DBA = \alpha$.

$\angle CBD = \alpha + 45^\circ$ (from (i) and (ii)).

But $\angle CXB = \alpha + 45^\circ$ (exterior angle of $\triangle AXC$ equals sum of interior opposite angles)

so $\angle CBD = \angle CXB$. $\square$

(iv) $\angle DEY = \alpha + 45^\circ$ from (ii) above,

$\angle CXB = \alpha + 45^\circ$ from (iii) above,

so $\angle DEY = \angle CXB$,

so $XYED$ is a cyclic quadrilateral as the exterior angle equals interior opposite angle. $\square$

QUESTION FOUR

(a) Prove that for positive integers n ,

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n! = (n+1)! - 1.$$

STEP 1.

When $n = 1$:

$$\text{LHS} = 1$$

$$\text{RHS} = 2! - 1$$

$$= 1.$$

So proposition is true for $n = 1$. $\square$

STEP 2.

Assume the proposition true for some positive integer k so that,

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + k \times k! = (k+1)! - 1. \quad \square$$

We are required to prove the proposition true for $k+1$. That is,

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + k \times k! + (k+1) \times (k+1)! = (k+2)! - 1.$$

$$\begin{aligned}
 \text{Now LHS} &= 1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k! + (k+1) \times (k+1)!. \\
 &= (k+1)! - 1 + (k+1) \times (k+1)!, \text{ from the assumption } \boxed{\checkmark} \\
 &= (k+1)! \times (1+k+1) - 1 \\
 &= (k+2)(k+1)! - 1 \\
 &= (k+2)! - 1 \\
 &= \text{RHS. } \boxed{\checkmark}
 \end{aligned}$$

It follows from steps one and two above by mathematical induction that for all positive integers n ,

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1.$$

(b) (i) $y^2 = -\frac{a}{2}(x-a).$

$$\begin{aligned}
 \text{LHS} &= y^2 \\
 &= a^2 \sin^2 \theta \\
 \text{RHS} &= -\frac{a}{2}(a \cos 2\theta - 1) \\
 &= -\frac{a^2}{2}(1 - 2\sin^2 \theta - 1) \boxed{\checkmark} \\
 &= a^2 \sin^2 \theta \\
 &= \text{LHS. } \boxed{\checkmark}
 \end{aligned}$$

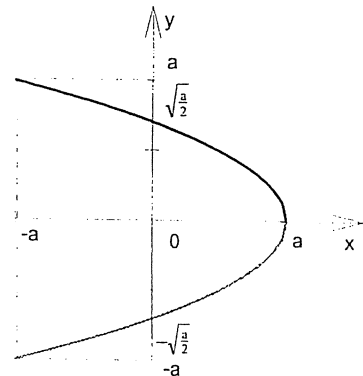
So P lies on the curve.

(ii) $x = a \cos 2\theta$

so $-a \leq x \leq a,$

and $y = a \sin \theta$

so $-a \leq y \leq a. \boxed{\checkmark}$



$$y^2 = -\frac{a}{2}(x-a). \boxed{\checkmark}$$

(c) (i) Given points are $A(-1, 3)$ and $(2, 8).$

$$\begin{aligned}
 \text{For } P \text{ } x &= \frac{k \times 2 + 1 \times -1}{k+1} \\
 &= \frac{2k-1}{k+1}
 \end{aligned}$$

$$\begin{aligned}
 \text{and } y &= \frac{k \times 8 + 1 \times 3}{k+1} \\
 &= \frac{8k+3}{k+1}.
 \end{aligned}$$

P is the point $\left(\frac{2k-1}{k+1}, \frac{8k+3}{k+1}\right). \boxed{\checkmark\checkmark}$

(ii) Let P lie on $5x + 2y - 10 = 0,$

$$\text{so } 5 \times \left(\frac{2k-1}{k+1} \right) + 2 \times \left(\frac{8k+3}{k+1} \right) - 10 = 0 \quad \boxed{\checkmark}$$

$$10k - 5 + 16k + 6 - 10k - 10 = 0$$

$$16k - 9 = 0$$

$$k = \frac{9}{16}. \quad \boxed{\checkmark}$$

The line $5x + 2y - 10 = 0$ divides the interval in the ratio 9 : 16.

QUESTION FIVE

(a) (i) $P(x) = (x+1)(x-2)Q(x) + R(x)$

$$P(-1) = R(-1)$$

$$\text{so } R(-1) = 3. \quad \boxed{\checkmark}$$

(ii) $P(2) = 0 + R(2)$

$$\text{so } R(2) = -2. \quad \boxed{\checkmark}$$

$$\text{Now } -a + b = 3$$

$$\text{and } 2a + b = -2$$

$$\text{so } 3a = -5$$

$$a = -\frac{5}{3} \quad \boxed{\checkmark}$$

$$\text{and } b + \frac{5}{3} = 3$$

$$b = \frac{4}{3}. \quad \boxed{\checkmark}$$

(b) (i) $t_r = {}^{12}C_r \times 2^{12-r} \times 3^3 \quad \boxed{\checkmark}$

$$t_{r+1} = {}^{12}C_{r+1} \times 2^{11-r} \times 3^{r+1}$$

$$\frac{t_{r+1}}{t_r} = \frac{{}^{12}C_{r+1} \times 2^{11-r} \times 3^{r+1}}{{}^{12}C_r \times 2^{12-r} \times 3^3}$$

$$= \frac{12!}{(r+1)!(11-r)!} \times \frac{r!(12-r)!}{12!} \times \frac{3}{2} \quad \boxed{\checkmark}$$

$$= \frac{12-r}{r+1} \times \frac{3}{2}$$

$$= \frac{36-3r}{2r+2} \quad \boxed{\checkmark}$$

(ii) For increasing coefficients,

$$\frac{t_{r+1}}{t_r} > 1$$

$$\frac{36-3r}{2r+2} > 1$$

$$36-3r > 2r+2$$

$$r < 6\frac{4}{5}.$$

Since $t_r < t_{r+1}$, $t_1 < t_2 < t_3 < \dots < t_7$.

So the greatest coefficient is $t_7 = {}^{12}C_7 \times 2^5 \times 3^7. \quad \boxed{\checkmark}$

$$(c) \quad (i) \quad (1+x)^n \times (1+x)^n = \left(\binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n \right)^2.$$

Term in x^n

$$= \binom{n}{0} \binom{n}{n} x^n + \binom{n}{1} \binom{n}{n-1} x^n + \cdots + \binom{n}{n} \binom{n}{0} x^n$$

$$= \left(\binom{n}{0} \binom{n}{n} + \binom{n}{1} \binom{n}{n-1} + \cdots + \binom{n}{n} \binom{n}{0} \right) x^n$$

Coefficient of term in x^n

$$= \binom{n}{0} \binom{n}{n} + \binom{n}{1} \binom{n}{n-1} + \cdots + \binom{n}{n} \binom{n}{0} \quad \boxed{\checkmark}$$

$$= \binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2, \text{ since } \binom{n}{r} = \binom{n}{n-r}$$

$$= \sum_{r=0}^n \binom{n}{r}^2. \quad \boxed{\checkmark}$$

(ii) For coefficient of term in x^n in the expansion of $(1+x)^{2n}$,

$$\text{coefficient} = \binom{2n}{n}$$

$$= \frac{(2n)!}{(n!)^2} \quad \boxed{\checkmark}$$

$$\text{so } \sum_{r=0}^n \binom{n}{r}^2 = \frac{(2n)!}{(n!)^2} \quad \boxed{\checkmark}$$

QUESTION SIX

$$(a) \quad (i) \quad f(x) = x^3 - 3x + 1$$

$$f'(x) = 3x^2 - 3.$$

$$\text{Let } f'(x) = 0$$

$$3x^2 - 3 = 0$$

$$3(x-1)(x+1) = 0$$

$$x = 1 \text{ or } -1. \quad \boxed{\checkmark}$$

$$(ii) \quad f(0) = 1$$

$$> 0$$

$$\text{and } f(0.5) = -0.375$$

$$< 0,$$

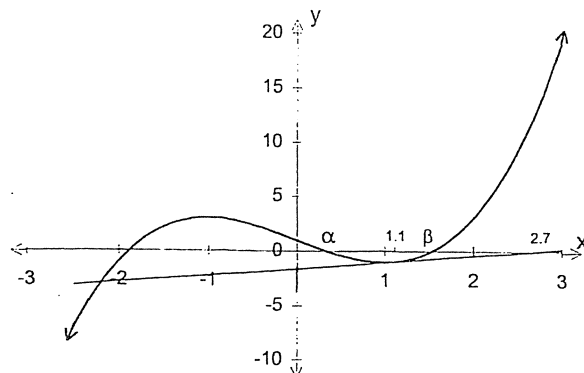
$$\text{so } \alpha \text{ is a root between } x = 0 \text{ and } x = 0.5. \quad \boxed{\checkmark}$$

$$(iii) \quad x_2 = 0.1 - \frac{f(0.1)}{f'(0.1)}$$

$$= 0.1 + \frac{0.701}{2.97} \quad \boxed{\checkmark}$$

$$= 0.336 \text{ (correct to three decimal places).} \quad \boxed{\checkmark}$$

(iv)



The tangent at $x = 1.1$ crosses the x -axis to the right of the root β shown, where $\alpha < \beta$. Further applications will approximate to the root at β . $\boxed{\checkmark}$

$$\begin{aligned} \text{(b) (i) Gradient} &= \frac{ap^2 - aq^2}{2ap - 2aq} \\ &= \frac{p + q}{2} \end{aligned}$$

Equation of the chord is,

$$y - ap^2 = \frac{p + q}{2}(x - 2ap)$$

$$y = \frac{p + q}{2}x - apq. \quad \boxed{\checkmark}$$

(ii) (α) $S(0, a)$ lies on the chord,

$$\text{so } a = 0 - apq$$

$$pq = -1.$$

PQ is parallel to x -axis so gradient is zero.

$$\frac{p + q}{2} = 0$$

$$\text{so } p + q = 0 \quad \boxed{\checkmark \text{ for both results}}$$

and so $p = 1$ and $q = -1$.

It follows that the coordinates are $P(2a, a)$ and $Q(-2a, a)$. $\boxed{\checkmark}$

$$(\beta) \quad y = \frac{1}{4a}x^2$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

At P , $\frac{dy}{dx} = 1$, so gradient of normal is -1 .

Equation of normal:

$$y - a = -1(x - 2a)$$

$$y = -x + 3a \dots (1)$$

$$\text{At } Q, \frac{dy}{dx} = -1, \text{ so gradient of normal is } 1. \quad \boxed{\checkmark \text{ for either gradient}}$$

Equation of normal:

$$y - a = 1(x + 2a)$$

$$y = x + 3a \dots (2) \quad \boxed{\checkmark \text{ for normals}}$$

For N , (1) + (2):

$$2y = 6a$$

so $y = 3a$

and $3a = x + 3a$

so $x = 0$.

N is the point $(0, 3a)$.

- (γ) Since the tangents to the parabola at P and Q are also tangents to the circle, the normals at these points are radii of the circle. These intersect at the centre of the circle.

The centre of the circle is $N(0, 3a)$.

$$\begin{aligned}\text{radius} &= \sqrt{(2a - 0)^2 + (3a - a)^2} \\ &= \sqrt{4a^2 + 4a^2} \\ &= 2a\sqrt{2}. \quad \boxed{\checkmark}\end{aligned}$$

The equation of the circle is:

$$\begin{aligned}(x - 0)^2 + (y - 3a)^2 &= (2a\sqrt{2})^2 \\ x^2 + y^2 - 6ay + 9a^2 &= 8a^2 \\ x^2 + y^2 - 6ay + a^2 &= 0. \quad \boxed{\checkmark}\end{aligned}$$

QUESTION SEVEN

(a) $\cos\left(2\pi\left(1 - \frac{1}{3}x\right)\right) = -\frac{1}{2}$

$$2\pi\left(1 - \frac{1}{3}x\right) = 2n\pi + \frac{2\pi}{3} \text{ or } 2n\pi - \frac{2\pi}{3}, \text{ where } n \text{ is an integer.} \quad \boxed{\checkmark}$$

$$1 - \frac{1}{3}x = n + \frac{1}{3} \text{ or } n - \frac{1}{3}$$

$$-\frac{1}{3}x = n - \frac{2}{3} \text{ or } n - \frac{4}{3}$$

$$x = 2 - 3n \text{ or } 4 - 3n \quad \boxed{\checkmark\checkmark}$$

(Note: Check variations.)

(b) (i)

$$V \cos \theta = \frac{x}{t}$$

$$V \sin \theta = \frac{y}{t} + \frac{1}{2}gt$$

$$\left(\frac{x}{t}\right)^2 + \left(\frac{y}{t} + \frac{1}{2}gt\right)^2 = V^2$$

$$x^2 + \left(y + \frac{1}{2}gt^2\right)^2 = V^2t^2$$

$$4x^2 + (2y + gt^2)^2 = 4V^2t^2$$

$$\text{so } 4y^2 + 4ygt^2 + (4x^2 + g^2t^4 - 4V^2t^2) = 0$$

$$4y^2 + 4gt^2y + k = 0 \quad \boxed{\checkmark}$$

(ii) Now $t = \frac{x}{V \cos \theta}$ which, for fixed x and V , is a minimum when $\cos \theta$ is a maximum.

So minimum $t = \frac{x}{V}$ when $\theta = 0$ $\square$

$$= \frac{10}{V} \cdot \square$$

So $y = -\frac{1}{2}g \times \left(\frac{10}{V}\right)^2$

$$= -\frac{50g}{V^2} \cdot \square$$

- (iii) For $t > \frac{10}{V}$ there is a solution for θ , which means that the projectile hits the wall and hence there will be a solution for y .

Now for the quadratic in y :

$$\Delta = (4gt^2)^2 - 16(g^2t^4 + 4 \times 10^2 - 4V^2t^2)$$

$$= 64(V^2t^2 - 100).$$

So $\Delta > 64\left(V^2 \times \frac{10^2}{V^2} - 100\right)$ since $t > \frac{10}{V}$

$$> 0. \quad \square$$

Hence there are two real and distinct roots and so there are two impacts at time t . $\square$

Now distance between impacts equals difference between roots.

$$\text{Difference} = \frac{-b + \sqrt{\Delta}}{2a} - \frac{-b - \sqrt{\Delta}}{2a}$$

$$= \frac{2\sqrt{\Delta}}{2a}$$

$$= \frac{\sqrt{\Delta}}{a}$$

So Distance $= \frac{8\sqrt{V^2t^2 - 100}}{4}$

$$= 2\sqrt{V^2t^2 - 100}. \quad \square$$

$$\begin{aligned} \text{(iv)} \quad \text{Distance} &= 2\sqrt{V^2t^2 - 100} \\ &= 2\sqrt{100t^2 - 100} \\ &= 20\sqrt{t^2 - 1} \end{aligned}$$

$$\begin{aligned} \text{so} \quad 20\sqrt{t^2 - 1} &= 20\sqrt{3} \\ \sqrt{t^2 - 1} &= \sqrt{3} \\ t^2 - 1 &= 3 \\ t^2 &= 4 \\ t &= 2. \quad \boxed{\checkmark} \end{aligned}$$

$$\begin{aligned} \text{Now} \quad \cos \theta &= \frac{x}{Vt} \\ &= \frac{10}{10 \times 2} \\ &= \frac{1}{2} \end{aligned}$$

and so $\theta = 60^\circ$ and -60° are the angles of elevation. $\boxed{\checkmark}$

GJ